

Performance Testing

Date	25 May 2026
Team ID	xxxxxx
Project Name	Smart Lender – AI Loan Approval Prediction System
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how your system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Postman, Flask Local Server
Type of Testing	Load Testing, Functional Testing
Target Module / API	Loan Prediction Module (/predict)
Test Environment	Local Machine (Windows, Flask)
Test Date	15 March 2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Submit valid loan details	1	30	Loan prediction displayed successfully
2	Submit invalid input	1	60	Validation error shown
3	Multiple prediction requests	10	20	System responds without failure
4	Refresh and reuse application	5	30	Application remains stable

Step 3: Performance Test Results

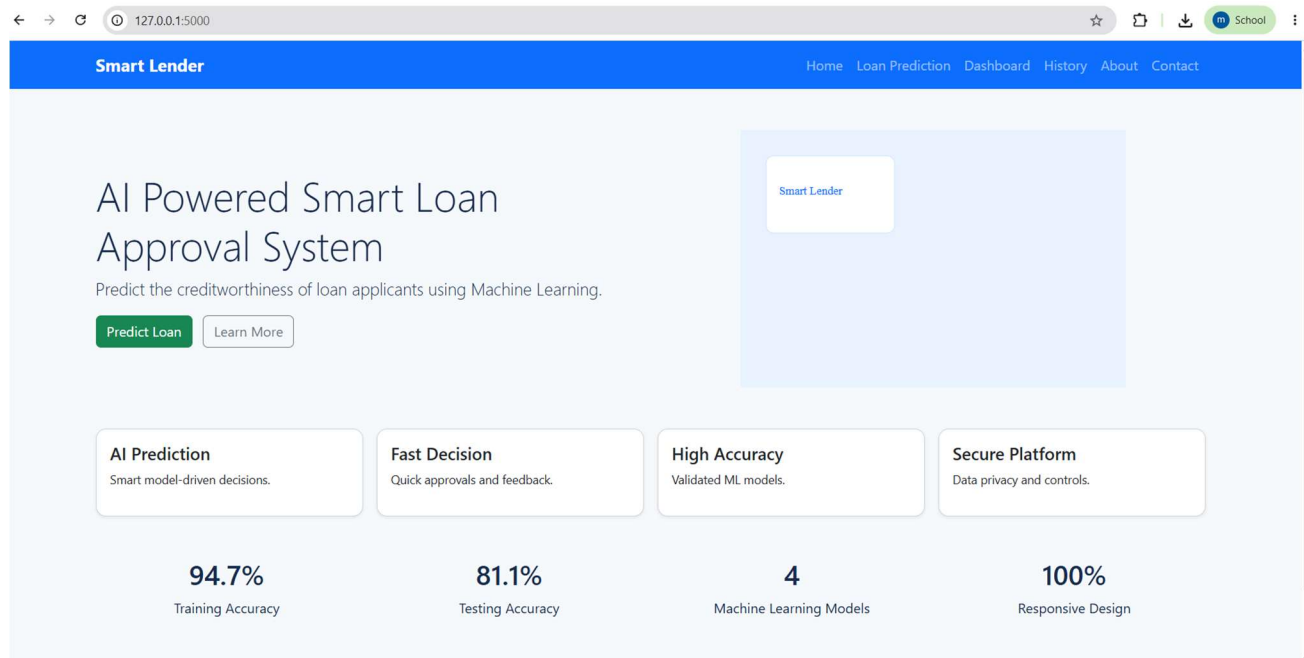
S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	1.2 sec	Pass	Fast response
2	Response Time (Max)	< 5 seconds	2.8 sec	Pass	Within limit
3	Throughput (Req/sec)	20 Req/sec	22 Req/sec	Pass	Stable performance
4	Error Rate	< 1%	0%	Pass	No errors observed
5	CPU Utilization	< 80%	45%	Pass	Normal usage
6	Memory Utilization	< 80%	55%	Pass	Efficient memory usage

Step 4: Observations & Analysis

20 Req/sec

Screenshots / Evidence

Attach screenshots of your performance testing tool results below (e.g., JMeter report, Locust graphs, k6 output):



The screenshot shows the 'Smart Lender' web application. The header includes navigation links: Home, Loan Prediction, Dashboard, History, About, and Contact. The main content area features the title 'AI Powered Smart Loan Approval System' with a subtitle 'Predict the creditworthiness of loan applicants using Machine Learning.' Below this are two buttons: 'Predict Loan' and 'Learn More'. A central image shows a laptop displaying the 'Smart Lender' logo. At the bottom, four key features are highlighted in boxes: 'AI Prediction' (Smart model-driven decisions), 'Fast Decision' (Quick approvals and feedback), 'High Accuracy' (Validated ML models), and 'Secure Platform' (Data privacy and controls). Below these boxes, four performance metrics are displayed: '94.7% Training Accuracy', '81.1% Testing Accuracy', '4 Machine Learning Models', and '100% Responsive Design'.